

N2S Eval Workbench — A Simple Tour

Evaluate → Deep dive → Intervene α → SAE features (downstream Track A + B)

CXR N2S lab notes (workbench on port 8257)

September 2026

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Who this is for

This guide explains **what we built and ran** in the N2S Eval Workbench, in plain language.

You do **not** need to already know mechanistic interpretability (MI) or Representation Engineering (RepEng). Those ideas are introduced briefly below, then tied to the **four** workbench actions:

Evaluate → Deep dive → Intervene α → SAE features

One sentence version of the project:

We paste a doctor's note, ask a model whether "first-line therapy failed," then optionally look *inside* the model, nudge a frozen direction, and (when needed) decompose that direction into sparse features — while Track A keeps AUTO vs REVIEW auditable.

Scope of this tour: the **downstream** path (utilization / generation) plus the **neural→symbolic boundary** (Track A Dual / gates).

Upstream (formation map, U-A/U-B, live Score on **:8258**) has its own complete write-up: docs/N2S-Upstream-Workbench-Tour.md / .pdf — including a full decode of $d = \text{unit}(\mu_T - \mu_C)$.

One-page claim freeze: lab docs/N2S-SYSTEM-CLAIM.md · closeout TRACKB-DOWNSTREAM-CLOSEOUT.md · upstream portfolio TRACKB-UPSTREAM-PORTFOLIO.md · Track A TRACKA-TEMPORAL-FAMILY-FREEZE.md · circuit TRACKB-CIRCUIT-C01-FREEZE.md.

System characterization (three pieces)

Doctor's note

↓ UPSTREAM – formation / evolution of representations (:8258 + lab)

Tokenization → layer-wise compute → internal reps

↓ DOWNSTREAM – utilization → structured LLM output (this tour · :8257)

Generation / commit (e.g. contradiction.present)

NEURAL → SYMBOLIC BOUNDARY (hard cut · Track A)

CXR atoms / Dual / gates → AUTO | REVIEW

Upstream vs downstream is an **experimental partition** (formation → utilization), not a hard layer line inside the transformer. The only hard cut is neural → symbolic.

Background theory (gentle)

Neural-to-symbolic (N2S)

Large language models are good at reading messy clinical text. Rules engines are good at **auditable** decisions. N2S means:

- 1. **Neural** — the model extracts structured claims (therapy, given?, failure?, conflict?).
- 2. **Symbolic** — deterministic code turns those claims into atoms and fires a rule.
- 3. **Gates** — if extractors disagree, or claims cannot be evidenced, we **REVIEW** instead of silently AUTOing a wrong answer.

So Track A is mostly **reliability engineering**, not “interpretability theater.”

What is MI?

Mechanistic interpretability asks: *what internal computations* produce a behavior?

In our workbench we take **small, causal steps** into MI on HF **Qwen 2.5 7B**:

- Read **activations** (hidden vectors) at selected **layers** (Deep dive).
- Ask whether **L20** looks more like a **temporal-change** or **contradiction** pattern.
- **Steer** L20 with a frozen direction **v** (Intervene α).
- **Decompose v** with a published L20 **SAE** and steer top-k features (SAE features).
- **Circuit ladder (lab CLI)**: who writes along **v** at commit (C0) → patch L20 MLP (C1) → path restrict (C2). Soft gates frozen — see Phase 5.

Stack: Hugging Face transformers + PyTorch **forward hooks** (callbacks on a layer’s output). We do **not** rewrite the model into TransformerLens for this workbench.

What is RepEng?

Representation Engineering (Zou et al.–style idea, applied narrowly here) says:

If a concept lives as a direction in activation space, you can **read** it (probe) and sometimes **write** it (steer).

Our write step is simple:

$$\mathbf{h} \leftarrow \mathbf{h} + \alpha \cdot \mathbf{v}$$

Glossary — how to read $\mathbf{v} = \text{unit}(\mu_A - \mu_B)$

Symbol	Plain English
Class A	Gold SATISFIED temporal-change notes where HF already said X=false (expand fit)
Class B	Gold CONTRADICTION notes where HF said X=true
μ_A / μ_B	Mean commit-token residual at L20 over those classes
$\mu_A - \mu_B$	Difference pointing toward “more A / less B” geometry
unit(·)	Normalize to length 1 — direction only
v	Frozen expand direction used by Intervene / SAE / circuit
α (alpha)	How hard we push; frozen claim uses α = 8
h	Hidden state at L20 on the contradiction.present commit token

So **v** points roughly from “true contradiction geometry” toward “temporal / non-contradiction geometry.” Pushing along **v** can clear some **false X** without being a magic family-wide fix.

Do not confuse with Upstream $d = \text{unit}(\mu_T - \mu_C)$ (prefill note means @ L24). $\cos(d, v) \approx 0.03$ — almost orthogonal. Full Upstream decode: N2S-Upstream-Workbench-Tour.pdf.

Why controls matter

Any steer can look impressive until you check:

Control	What it rules out
baseline	“Did X already flip without us?”
gaussian_unit	“Would <i>any</i> random push of size alpha work?”
reverse_vector (-v)	“Is the sign of v meaningful?”

A credible story: **full_vector** flips false X; gaussian does not; reverse does not help; true contradictions stay X=true.

Two clinical traps (memorize these)

1. **True contradiction** — same visit says “failed” *and* “still responding.” → gold **CONTRADICTION**.
2. **Temporal change** — responded *earlier*, progressed *later*. → gold often **SATISFIED** (therapy failed), **not** contradiction. Models sometimes wrongly set **X=true** (false contradiction).

Track A contains false-X via grounding + Dual + REVIEW. Track B asks whether L20 geometry *causes* some of that false X.

The clinical question (Track A)

We care about one formal question (a **predicate**):

Did **first-line therapy fail**?

In code: FIRST_LINE_THERAPY_FAILED. Roughly:

Atom	Meaning
A	A first-line therapy is identified
B	It was actually given (not only planned / declined)
C	A failure / progression event happened
D	That failure is of that first-line
X	Same-time incompatible claims → short-circuit CONTRADICTION

Rule sketch: if X → CONTRADICTION; else if $A \wedge B \wedge C \wedge D \rightarrow$ SATISFIED; else NOT_SATISFIED / UNCERTAIN.

REVIEW means a human should look — Dual paths disagreed, evidence gate failed, verify failed, or neural X $\neq$ grounded X (**N2S mismatch**).

The workbench in four buttons

Run at <http://127.0.0.1:8257/> (prefer `../cxrlabs/faiss_gpu1/bin/python server.py` for HF GPU phases).

Button	What it does	Model used	Enabled when
Evaluate	Fast Track A: extract → ground → Dual → AUTO / REVIEW	Ollama (e.g. Qwen 32B)	Always (with a note)
Deep dive	Round-trip + L20 readout (forensics)	HF Qwen 7B	After Evaluate
Intervene α	Steer L20 with $\mathbf{v}$; baseline + full_vector (+ optional gaussian / reverse)	HF Qwen 7B	Note pasted (Evaluate optional)
SAE features	Rank Chanin L20 sparse features vs $\mathbf{v}$; steer top-k ($\pm$ / random / group)	HF Qwen 7B + SAE weights	Note pasted (Evaluate optional)

Rule of thumb: Evaluate first for Track A. Deep dive to *observe* L20. Intervene for a one-direction *causal* nudge. SAE when you need a *sparser* decomposition of the same $\mathbf{v}$.

UI hygiene (shipped): hard-refresh restores the last Evaluate / Deep dive / Intervene / SAE panel from **sessionStorage** in this browser tab (note + results). Intervene and SAE stay available whenever a note is present (they do not require Deep dive).

Built-in UI pastes: **EX_BC_E1**, **EX_CONTRA**, **EX_PLANNED**. Harder DEV fixtures (TX_E14, TF_E5, ...) live in the lab data/ JSON files.

Functions at play (map)

Think of three layers: **UI API** → **Track A library** → **Track B library**.

Workbench API (cxr-n2s-eval-workbench/)

Function	Role
evaluate_fast	One-shot Track A Evaluate
submit_deep_dive / _deep_dive_work	Async Deep dive job
submit_intervene	Async Intervene job → lab
run_intervene_live	
submit_sae	Async SAE job → lab
run_workbench_sae	
job_status / meta_payload	Poll progress; ship example notes to UI
_verified_pipeline	extract → ground → verify → rule (per Dual path)
_mismatch	neural X vs grounded X → REVIEW signal

HTTP routes (port **8257**): POST /api/evaluate, /api/deep-dive, /api/intervene, /api/sae; GET /api/job, /api/meta.

Track A lab (cxr-evidence-grounding-lab/n2s_lab/)

Function	Module	Plain role
analyze_free_text	neural.py	Short free-text read of the note (feeds path D)
extract_live	neural.py	Note → structured Extraction (path C)
extract_from_analysis	neural.py	Analysis text → Extraction (path D)
extract*_repair	neural.py	One repair pass after verify fail
paraphrase_from_extraction	neural.py	Structure → paraphrase (round-trip)
ground	ground.py	Extraction → atoms A-D / X; clears false X on sequenced temporal change
verify_formalization	verify.py	Did structure keep important distinctions?
evaluate_rule	predicate.py	Atoms → SATISFIED / NOT_SATISFIED / CONTRADICTION / ...
dual_path_verdict	roundtrip.py	C and D agree → AUTO candidate; else REVIEW
verify_roundtrip	roundtrip.py	Paraphrase must preserve conflict / uncertainty
gate_verdict_with_evidence	evidence.py	Claims must be spannable in the note
apply_gates / classify_case	auto_contract.py	Final AUTO vs REVIEW; wrong_AUTO buckets

Evaluate wiring order: analyze → path C & D pipelines → evidence gate → Dual → N2S mismatch → gates → AUTO/REVIEW.

Track B lab (same package)

Function	Module	Plain role
build_directions_from_expand	n2s_forensics.py	Build L20 unit(mean_A - mean_B) artifact
load_directions	n2s_forensics.py	Load frozen directions JSON
score_layer_hiddens	n2s_forensics.py	Soft scores temporal vs contradiction @ layers 8-24
run_forensics_live	n2s_forensics.py	HF 7B forward + L20 readout
format_forensics_text	n2s_forensics.py	Human-readable forensics panel
load_expand_direction	n2s_intervene.py	Load expand $\mathbf{v}$ (not old Phase-10 BC_E* vector)
run_intervene_live	n2s_intervene.py	Four arms @ alpha (default 8)
format_intervene_text	n2s_intervene.py	Human-readable intervene panel
run_workbench_sae / pilot CLI	n2s_sae_pilot.py	Load Chanin L20 SAE; rank feats vs $\mathbf{v}$; top-k steers

Direction artifact: artifacts/n2s-forensics-directions-qwen7b.json.

SAE pilot docs: lab docs/TRACKB-SAE-PILOT.md.

Frozen claim write-up: lab docs/TRACKB-ALPHA8-FREEZE.md.

Phase 1 — Evaluate

What Evaluate is

Paste a note → two formalization paths:

- **C** — note → structure
- **D** — free-text analysis → structure

If they **agree**, Dual can **AUTO**.

If they **disagree**, Dual stays **REVIEW**. That disagreement is a *feature*: we do not force every note to AUTO with one-off patches.

N2S mismatch: extractor claims contradiction (**neural X**) while grounding cleared it (**grounded X**) → **REVIEW**.

Example A — true contradiction → AUTO CONTRADICTION

Workbench id: EX_CONTRA

Note A: Patient failed first-line FOLFOX after four cycles. Addendum same day: Disease remains responsive to FOLFOX; continue current regimen.

Wanted: CONTRADICTION.

Evaluate: C = D = Dual = CONTRADICTION; neural X and grounded X agree → **AUTO**.

The screenshot displays the N2S Eval Workbench interface. On the left, the 'CLINICAL NOTE' section contains the text: 'Note A: Patient failed first-line FOLFOX after four cycles. Addendum same day: Disease remains responsive to FOLFOX; continue current regimen.' Below this, there are buttons for 'Evaluate', 'Deep dive', 'Download JSON', and 'Download report'. The 'Evaluate' button is highlighted. On the right, the 'RESULT' section shows 'AUTO CONTRADICTION' with a green 'AUTO' label. Below this, it states 'AUTO: CONTRADICTION' and 'model qwen2.5-coder:32b · case LIVE_85ccdb87 · live true'. Further down, two panels show 'C_FULL (NOTE → STRUCTURE)' and 'D_FULL (ANALYSIS → STRUCTURE)', both displaying 'CONTRADICTION'. Below these, the 'DUAL_FULL' section shows 'CONTRADICTION' with 'paths_agree'. The 'N2S MISMATCH (EXTRACT X VS GROUNDED X)' section shows 'neural=true · grounded=true · agree'. The 'GATES / REASONS' section shows 'No gate failures'. The 'GROUNDING (C)' section shows a JSON object: { "first_line_identified": "true", "first_line_administered": "true", "failure_event": "true", "failure_of_first_line": "true", "contradiction": true, "uncertainty": false, ... }.

Figure 1: Evaluate: AUTO CONTRADICTION on same-day FOLFOX conflict

Example B — temporal failure with extract/ground fight → REVIEW

Live paste (response, then later progression):

Metastatic colorectal cancer. First-line FOLFOX produced a partial response. At follow-up four months later, imaging demonstrated new hepatic lesions... FOLFOX was discontinued for treatment failure...

Wanted clinically: SATISFIED (failure after prior response), **not** contradiction.

Evaluate: Dual may say SATISFIED, but extractor often sets X=true while grounding clears X as temporal-change → **N2S mismatch** → **REVIEW**.

The screenshot displays the N2S Eval Workbench interface. On the left, the 'CLINICAL NOTE' section contains the text: 'Metastatic colorectal cancer. First-line FOLFOX produced a partial response. At follow-up four months later, imaging demonstrated new hepatic lesions and the oncologist documented progression on FOLFOX. FOLFOX was discontinued for treatment failure and second-line FOLFIRI was initiated.' Below this, the 'Evaluate' button is highlighted. The 'RESULT' section on the right shows 'REVIEW' with a 'REVIEW' button and the text 'REVIEW: REVIEW · N2S mismatch'. Below this, the 'C_FULL (NOTE → STRUCTURE)' and 'D_FULL (ANALYSIS → STRUCTURE)' sections both show 'SATISFIED'. The 'DUAL_FULL' section shows 'SATISFIED' with 'paths_agree'. The 'N2S MISMATCH (EXTRACT X VS GROUNDED X)' section shows 'neural=true · grounded=false · DISAGREE' and a detailed explanation of the mismatch. The 'GATES / REASONS' section lists 'n2s_mismatch_extract_vs_ground_X'. The 'GROUNDING (C)' section shows a JSON object with 'first_line_identified': 'true', 'first_line_administered': 'true', 'failure_event': 'true', and 'failure_of_first_line': 'true'.

Figure 2: Evaluate: REVIEW with N2S mismatch on temporal note

This is the note that later becomes the Intervene “selective flip” demo.

Example C — clean temporal note already clean → AUTO SATISFIED

Workbench id: EX_BC_E1 (pembrolizumab + chemo, response → later progression).

Paths and X already agree → **AUTO SATISFIED**. Nothing broken; Evaluate is done. Soft cases like this are **poor** Intervene tests (baseline X often already false on 7B).

Example D — planned only → NOT_SATISFIED

Workbench id: EX_PLANNED

Newly diagnosed metastatic colon cancer. Multidisciplinary plan is to start first-line FOLFOX next week. No systemic therapy has been administered yet.

Wanted: NOT_SATISFIED (atom **B** false — therapy not given).

Evaluate: typically AUTO **NOT_SATISFIED** when Dual agrees. Deep dive comparison is **NOT EVALUATED** (commitment outside temporal-vs-contradiction probe space).

Example E — lab fixtures worth knowing

Id	Short story	Gold intent
BC_E1	R-CHOP response → later progression	SATISFIED (temporal)
TF_E5	FOLFOX stable @ c3 → mets @ c6	SATISFIED (temporal)
TF_C1	“never chemo” vs same-day “completed R-CHOP”	CONTRADICTION
TX_E14	BEP markers clear early → later node / AFP rise	SATISFIED; harder expand case
TX_E01	gem/nab response → later peritoneal implants	SATISFIED

TX_E14 / TX_E01 are better alpha stress-tests than soft BC_E1-style notes when you want baseline X=true on 7B.

Evaluate summary

Situation	Typical Evaluate result
Same-day fail vs respond	AUTO CONTRADICTION
Temporal fail, extractor over-claims X	REVIEW (containment)
Clean temporal	AUTO SATISFIED
Planned / declined / never given	AUTO NOT_SATISFIED

Phase 2 — Deep dive

What Deep dive is

After Evaluate, **Deep dive** adds:

- 1. **Round-trip** — paraphrase from structure; distinctions must survive (verify_roundtrip).
- 2. **N2S forensics** — HF **Qwen 2.5 7B** hidden states at layers 8 / 12 / 16 / **20** / 24.

At L20 we score proximity to two frozen poles (soft scores ~sum to 1):

- **Temporal-change** pole
- **Contradiction** pole

Those scores are **not** the Track A verdict. They are a **geometry readout**.

Important label rule

The probe only lives in **temporal-change vs contradiction**.
If commitment is **SATISFIED** or **NOT_SATISFIED**, we show:

Representation-commitment comparison: **NOT EVALUATED**

We only talk “agree / mismatch” when commitment is **CONTRADICTION**.

Example — true contradiction, probe agrees

Commitment CONTRADICTION; L20 leans contradiction → comparison can show agree.

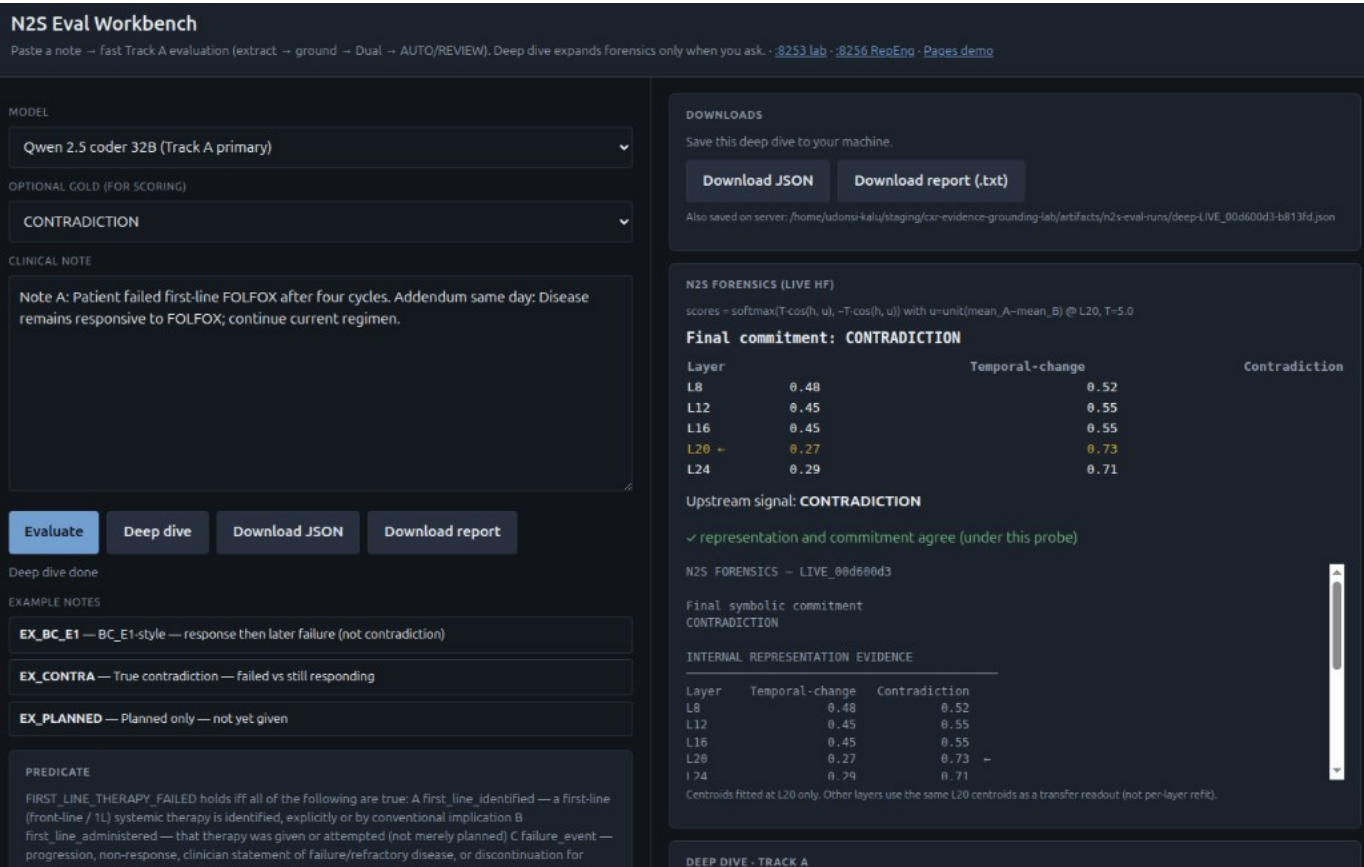


Figure 4: Deep dive: L20 leans contradiction; commitment CONTRADICTION

Example — NOT_SATISFIED (planned / declined), comparison not evaluated

Commitment NOT_SATISFIED; L20 may lean the non-contradiction pole. That is **not** a mismatch banner — comparison is **NOT EVALUATED**.

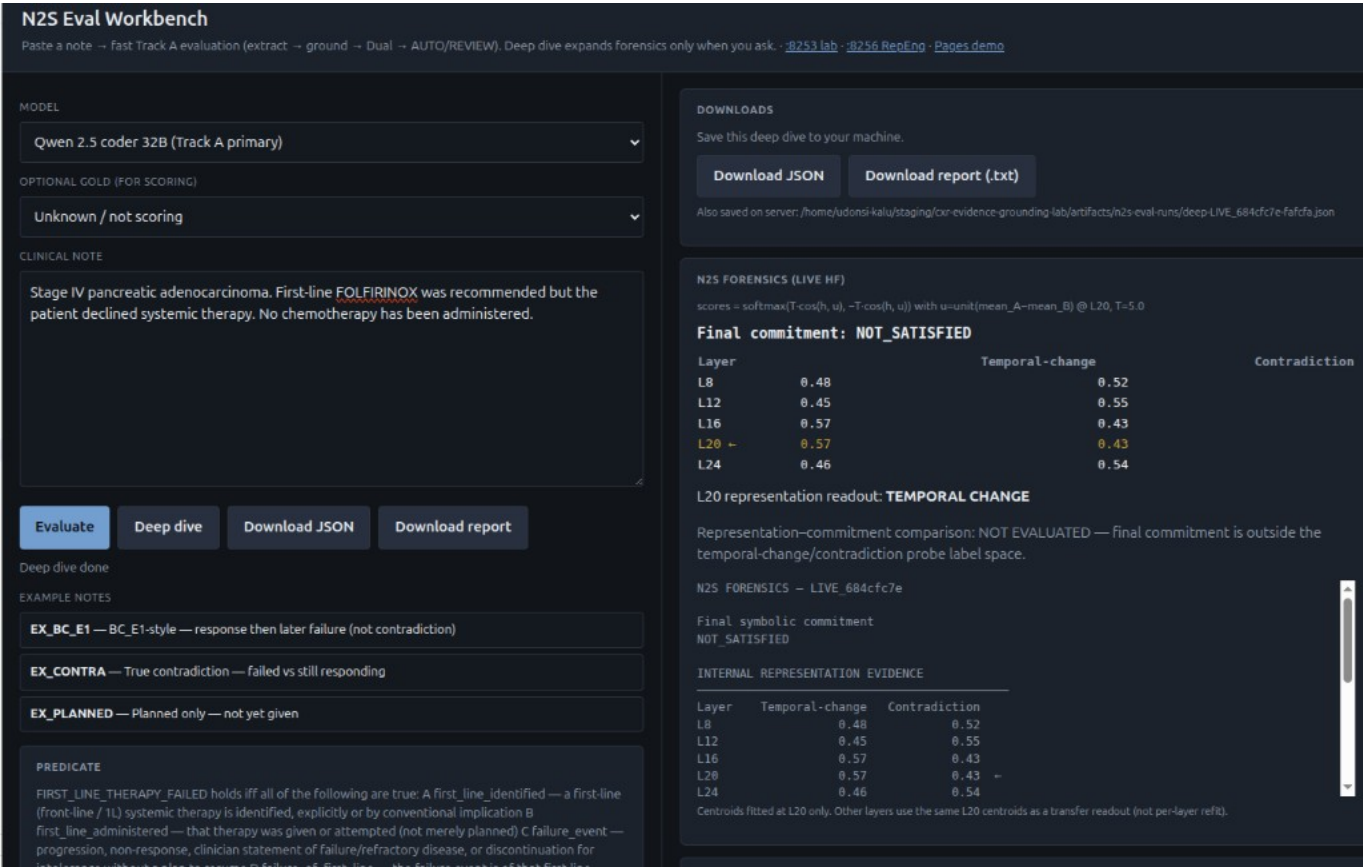


Figure 5: Deep dive: NOT_SATISFIED → comparison NOT EVALUATED

Example — temporal REVIEW note

On the FOLFOX temporal paste that Evaluate flagged REVIEW: Deep dive still **observes**. You may see L20 lean one pole while Track A stays REVIEW because of Dual / N2S mismatch. Deep dive does **not** override Track A.

Deep dive summary

Piece	Role
Track A Dual	Symbolic decision / AUTO vs REVIEW
L20 readout	Soft probe: temporal vs contradiction geometry
Comparison	Only meaningful when commitment = CONTRADICTION

Deep dive **observes**. It does not change the model.

Phase 3 — Intervene (alpha)

What Intervene is

Take the frozen expand direction **v**, add **alpha · v** at L20 on the contradiction-commit token, and measure whether `contradiction.present` flips.

- Default **alpha = 8** (frozen limited claim)
- **Not** the older Phase-10 BC_E* vector from port :8256
- Partial editor: historically flipped **3/8** weak-margin false-X fails on expand DEV; preserves contradiction + nofail controls
- Do **not** claim alpha=16/32 as the freeze (specificity / control damage)

The four arms (always read the table)

Arm	Meaning
baseline	No steer
full_vector	Steer with real v at alpha
gaussian_unit	Random direction (specificity control)
reverse_vector	Steer with -v (sign check)

Good false-X story: baseline X=true → full_vector X=false; gaussian stays true; reverse does not help.

Good true-contra story: all arms stay X=true.

Example — true contradiction: no flip (control)

Same-day FOLFOX (EX_CONTRA). Intervene alpha=8: baseline and full_vector both X=true. **No true→false flip.** Desired.

Example — temporal false-X: selective flip (the claim)

Temporal FOLFOX response→later progression. Track A **REVIEW** (N2S mismatch). HF 7B intervene:

Arm	X	Margin (approx.)
baseline	true	1.63
full_vector	false	~0
gaussian_unit	true	1.63
reverse_vector	true	3.50

That is the live workbench demonstration of the **limited alpha=8 expand editor**.

Example — when Intervene is a dull demo (still useful)

On **EX_BC_E1**-style notes, 7B baseline may already report X=false. Then full_vector has nothing dramatic to flip. That is fine: it means Track A + the smaller model already agree. Prefer **TX_E14 / TX_E01 / live temporal REVIEW** notes when teaching the causal claim.

N2S Eval Workbench
 Paste a note → Fast Track A evaluation (extract → ground → Dual → AUTO/REVIEW). Deep dive expands forensics only when you ask. · [B253 lab](#) · [B256 RepEng](#) · [Pages demo](#)

MODEL
 Qwen 2.5 coder 32B (Track A primary)

OPTIONAL GOLD (FOR SCORING)
 CONTRADICTION

CLINICAL NOTE
 Note A: Patient failed first-line FOLFOX after four cycles. Addendum same day: Disease remains responsive to FOLFOX; continue current regimen.

Evaluate Deep dive Intervene α Download JSON

Download report

INTERVENE A (HF 7B · EXPAND L20)
 $\alpha = 8$ (frozen claim)

☒ INCLUDE GAUSSIAN + REVERSE CONTROLS
 Limited α -B expand editor — not Ph10 BC_E* vector. Needs ~14 GB free after Ollama unload.

Intervene done

EXAMPLE NOTES
 EX_BC_E1 — BC_E1-style — response then later failure (not contradiction)
 EX_CONTRA — True contradiction — failed vs still responding

INTERVENE A= α · L20
 Frozen expand L20 editor (unit(mean_A-mean_B)); limited α =8 claim — partial editor, not family-wide fix. See docs/TRACKB-ALPHAB-FREEZE.md.

Arm	X	verdict	margin
baseline	true	CONTRADICTION	15.63
full_vector	true	CONTRADICTION	12.25
gaussian_unit	true	CONTRADICTION	15.38
reverse_vector	true	CONTRADICTION	18.25

No true → false X flip at this α (partial editor — expected often).

N2S INTERVENE — LIVE_76f022b3
 model: Qwen/Qwen2.5-7B-Instruct
 layer: L20 · $\alpha=8.0$

Frozen expand L20 editor (unit(mean_A-mean_B)); limited α =8 claim — partial editor, not family-wide fix. See docs/TRACKB-ALPHAB-FREEZE.md.

Arm	X	verdict	margin
baseline	True	CONTRADICTION	15.62
full_vector	True	CONTRADICTION	12.25
gaussian_unit	True	CONTRADICTION	15.38
reverse_vector	True	CONTRADICTION	18.25

artifact: /home/ludonis-kalu/staging/cxr-evidence-grounding-lab/artifacts/n2s-eval-runs/intervene-LIVE_76f022b3-f1fb6b.json

Download intervene JSON Download intervene report

RESULT
 AUTO **CONTRADICTION**
 AUTO: CONTRADICTION
 model qwen2.5-coder:32b · case LIVE_76f022b3 · live true

C_FULL (NOTE — STRUCTURE)
CONTRADICTION

D_FULL (ANALYSIS — STRUCTURE)
CONTRADICTION

Figure 6: Intervene: true contradiction — no X flip at $\alpha=8$

MODEL

Qwen 2.5 coder 32B (Track A primary)

OPTIONAL GOLD (FOR SCORING)

SATISFIED

CLINICAL NOTE

Metastatic colorectal cancer. First-line FOLFOX produced a partial response. At follow-up four months later, imaging demonstrated new hepatic lesions and the oncologist documented progression on FOLFOX. FOLFOX was discontinued for treatment failure and second-line FOLFIRI was initiated.

EvaluateDeep diveIntervene aDownload JSON

Download report

INTERVENE A (HF 7B - EXPAND L20)

a = B (frozen claim)

☒ INCLUDE GAUSSIAN + REVERSE CONTROLS

Limited a-B expand editor — not Phi10 BC_E* vector. Needs ~14 GiB free after Ollama unload.

Intervene done

EXAMPLE NOTES

EX_BC_E1 — BC_E1-style — response then later failure (not contradiction)

EX_CONTRA — True contradiction — failed vs still responding

EX_PLANNED — Planned only — not yet given

HF INTERVENE ARMS - A=B - L20

Frozen expand L20 editor (unit(mean_A-mean_B)); limited a-B claim — partial editor, not family-wide fix. See docs/TRACKB-ALPHAB-FREEZE.md.

Arm	X	verdict	margin
baseline	true	SATISFIED	1.63
full_vector	false	SATISFIED	-0.13
gaussian_unit	true	SATISFIED	1.63
reverse_vector	true	SATISFIED	3.50

full_vector flipped repair X true → false

N2S INTERVENE — LIVE_aa7c757b

model: Qwen/Qwen2.5-7B-Instruct

layer: L20 · a=8.0

Frozen expand L20 editor (unit(mean_A-mean_B)); limited a-B claim — partial editor, not family-wide fix. See docs/TRACKB-ALPHAB-FREEZE.md.

Arm	X	verdict	margin
baseline	True	SATISFIED	1.62
full_vector	False	SATISFIED	-0.12
gaussian_unit	True	SATISFIED	1.62
reverse vector	True	SATISFIED	3.50

artifact: /home/udoni-kalu/staging/cxr-evidence-grounding-lab/artifacts/n2s-eval-runs/intervene-LIVE_aa7c757b-0f6d43.json

Download intervene JSONDownload intervene report

RESULT

REVIEW REVIEW

REVIEW: REVIEW · N2S mismatch

model qwen2.5-coder:32b · case LIVE_aa7c757b · live true

C_FULL (NOTE — STRUCTURE)

SATISFIED

D_FULL (ANALYSIS — STRUCTURE)

SATISFIED

DUAL_FULL

SATISFIED

Figure 7: Intervene: temporal note — full_vector flips X true→false; controls do not

Intervene summary

Question	Answer from our runs
Does alpha=8 wreck true contradictions?	No (FOLFOX same-day)
Can alpha=8 clear some temporal false X?	Yes (selective flip + controls)
Does every temporal note need alpha?	No — many already X=false on 7B / AUTO on 32B
Is this SAE / TransformerLens?	No — one-direction steer. SAE is the next button (Phase 4), still HF hooks

Phase 4 — SAE features

What SAE features is

A **sparse autoencoder (SAE)** turns a dense residual vector into many mostly-off features. We reuse a **published** SAE for Qwen2.5-7B **layer 20**:

chanind/qwen2.5-7B-it-layer-20-saes (JumpReLU; residual post at block 20).

We do **not** train a new SAE for this pilot. Weights load via safetensors in n2s_sae_pilot.py (no sae_lens required).

Quest: decompose the same frozen expand direction **v** into sparse feature ids that (a) distinguish Class A vs B, (b) align with **v**, (c) track X / margin, then (d) **causally** steer top-k features on a pasted note — with a **true-contradiction control** (EX_CONTRA).

What the button runs

1. Rank features ($\Delta(A-B)$, cosine to **v**, correlation with X / margin).
 2. Steer top-k (UI default **3**) at strength **α** (same family as Intervene; often 8):
 - single-feature $\pm$
 - random-feature control
 - top-k group
 3. Optionally include **EX_CONTRA** so you can see whether sparsity wrecks true contradictions.
- HTTP: POST /api/sae (async job, same poll pattern as Intervene). Slow — many HF forwards.

How to read the panel

Column / block	Meaning
feat_id	Sparse feature index (no English dictionary)
$\Delta(A-B)$	Activation difference between expand classes
cos_v	Alignment with frozen expand v
r_X / r_margin	Tracks commit X / logit margin (when scored)
Arms table	baseline vs feature steers on your note (+ control note)

Claim hygiene (say / do not say)

Say	Do not say
We ranked L20 SAE features against frozen v and ran top-k steers	“We found the temporality neuron”
Ranking $\neq$ causal proof; arms table is the causal check	SAE replaces Dual / Track A
On soft notes where 7B baseline X=false , steers may look dull — still valid	Every FOLFOX temporal paste will show a dramatic flip

Say	Do not say
Control: true contra should stay X=true	Top-k group success $\Rightarrow$ production editor

Lab write-up: `cxr-evidence-grounding-lab/docs/TRACKB-SAE-PILOT.md`.
 Artifacts: `n2s-sae-pilot-scores.json`, `n2s-sae-pilot-causal*.json`.

SAE summary

Question	Answer from the pilot
New SAE trained?	No — Chanin L20 reused
Same v as Intervene?	Yes
English feature labels?	No
Workbench button?	Yes — SAE features on :8257
Interpreter?	Prefer faiss_gpu1 (same as other HF phases)

Phase 5 — Circuit ladder (lab CLI · frozen)

Not a fifth button on :8257 yet. After SAE, we asked: **who writes into the residual along v at commit?**

Step	Name	Result
C0	Write audit @ commit	Top $ \Delta(A-B) = \text{L20/mlp/frozen_v} \approx +6.75$
C1	Causal patch L20 MLP	Soft gate YES — $B \leftarrow A$ $\Delta\text{margin} \approx -3.1$; $A \leftarrow B$ flips X both
C2	Path restrict (L16 vs L20)	LOCAL_SUFFICIENT — L16 alone weak; stack $\approx$ L20

Say: L20 MLP commit write is a **causal site** (local at this grain).

Do not say: full temporality circuit / named head.

```
cd cxr-evidence-grounding-lab
./scripts/run_circuit_pilot.sh audit --max-per-class 2
./scripts/run_circuit_pilot.sh patch --max-per-class 2
./scripts/run_circuit_pilot.sh path --max-per-class 2
```

Freeze: TRACKB-CIRCUIT-C01-FREEZE.md. Downstream C0-C2 does **not** reopen Upstream U-C/U-D.

How the four phases fit together

Doctor's note

|
v

EVALUATE (Track A · downstream symbolic)

What did Track A output? AUTO or REVIEW?

Functions: extract, ground, verify, Dual, evidence, gates, mismatch

|
v

DEEP DIVE (observe L20)

What does L20 look like?

Functions: paraphrase/round-trip, run_forensics_live, score_layer_hiddens

|
v

INTERVENE α (write along v)

If we nudge L20 with v , does X flip?

Functions: load_expand_direction, run_intervene_live (+ controls)

|
v

SAE features (decompose v)

Which sparse feats align with v , and do top-k steers move X?

Functions: run_workbench_sae / n2s_sae_pilot

Product philosophy: When C and D disagree, **REVIEW is OK**. Investigate why; do not invent a special grounding hack for every new paste.

Ops notes:

- Evaluate may park a large Ollama model in VRAM. Deep dive / Intervene / SAE need ~14 GiB free for HF 7B — the workbench unloads Ollama and waits until /api/ps is clear before loading HF.
- Prefer ../cxrlabs/faiss_gpu1/bin/python server.py (not bare system python3) for GPU HF phases.
- Browser **sessionStorage** keeps the last results across hard-refresh in the same tab.

Not on :8257: Upstream live Score / Frozen replay / custom pair — use **:8258** (complete tour: N2S-Upstream-Workbench-Tour.pdf). Lab U1 cue $\pm\alpha\cdot v$ was a **null** (no commit X flip).

Frozen claims (visitor crib)

Stream	Allowed claim	Forbidden
Track A	Qwen Dual wrong_AUTO=0 on temporal DEV + sealed TEST; fails → REVIEW	Clinical validation; high AUTO coverage
Downstream $\alpha=8$	Partial L20 expand editor; controls held	Fixes all temporal false-X; $\alpha=16/32$ freeze
SAE	Chanin L20 decompose v + top-k steers (pilot)	“Found the temporality neuron”
Circuit C0-C2 Upstream	L20 MLP causal site ; C2 local-sufficient Formation d @ L24; gen YES; ablate + $\alpha \cdot d$ null ; :8258 Score correlational	Full circuit / temporality neuron Upstream editor; U-C/U-D from nulls

Full freeze text: [cxr-evidence-grounding-lab/docs/N2S-SYSTEM-CLAIM.md](#).

Upstream complete tour (including $d = \text{unit}(\mu_T - \mu_C)$ decode): [docs/N2S-Upstream-Workbench-Tour.pdf](#).

Tiny glossary

Term	Plain meaning
Track A	Reliability path: extract, ground, Dual, AUTO/REVIEW
Track B / MI / RepEng	Look inside / steer open-weight models
Upstream	How note reps form in prefill; map d @ L24; Score on :8258; ablate/ $\alpha \cdot d$ null
Downstream (this tour)	How internal reps are used for generation / structured output
Boundary	Neural structured output → CXR symbols / Dual / AUTO
Dual	Two formalization paths (C and D) must agree to AUTO
N2S	Neural extract → symbolic grounding
X	Contradiction flag (contradiction.present)
L20 / L24	Layer indices (Downstream steer/SAE @ L20; Upstream map peak @ L24)
v	$\text{unit}(\mu_A - \mu_B)$ — Downstream commit editor direction
d	$\text{unit}(\mu_T - \mu_C)$ — Upstream formation direction (see Upstream tour)
alpha	Steer strength along v (freeze = 8)
N2S mismatch	Neural X true, grounded X false → REVIEW
HF hooks	PyTorch register_forward_hook on HF layers — read/modify activations
SAE	Sparse autoencoder — Chanin L20 features over residual (Phase 4)
TransformerLens	Alternate hook/framework stack — study material; not required for :8257

Where the code lives

Piece	Path
Workbench UI / API	cxr-n2s-eval-workbench/ (port 8257)
Lab library	cxr-evidence-grounding-lab/n2s_lab/
Forensics	n2s_lab/n2s_forensics.py
Intervene	n2s_lab/n2s_intervene.py
SAE pilot	n2s_lab/n2s_sae_pilot.py
Direction artifact	artifacts/n2s-forensics-directions-qwen7b.json
SAE artifacts	artifacts/n2s-sae-pilot-scores.json, n2s-sae-pilot-causal*.json
alpha=8 freeze note	cxr-evidence-grounding-lab/docs/TRACKB-ALPHA8-FREEZE.md
SAE pilot note	cxr-evidence-grounding-lab/docs/TRACKB-SAE-PILOT.md
System claim (one-pager)	cxr-evidence-grounding-lab/docs/N2S-SYSTEM-CLAIM.md
Upstream complete tour	docs/N2S-Upstream-Workbench-Tour.md / .pdf (:8258)
Upstream server	upstream_server.py · upstream_api.py · static/upstream.html
Circuit freeze	cxr-evidence-grounding-lab/docs/TRACKB-CIRCUIT-C01-FREEZE.md
Upstream portfolio	cxr-evidence-grounding-lab/docs/TRACKB-UPSTREAM-PORTFOLIO.md
Upstream prefill	cxr-evidence-grounding-lab/docs/TRACKB-UPSTREAM-PREFILL.md
Track A temporal freeze	cxr-evidence-grounding-lab/docs/TRACKA-TEMPORAL-FAMILY-FREEZE.md
DEV fixtures	data/temporal-family-dev.json, temporal-family-dev-expand.json

Do **not** fit on sealed temporal-family-test.json.

Rebuild this PDF: ./build-newcomer-tour-pdf.sh → docs/N2S-Workbench-Newcomer-Tour.pdf.

Rebuild Upstream PDF: ./build-upstream-tour-pdf.sh → docs/N2S-Upstream-Workbench-Tour.pdf.

Bottom line

1. **Evaluate** decides AUTO vs REVIEW for the clinical predicate using extract → ground → Dual → gates (**boundary**).
2. **Deep dive** shows an L20 temporal-vs-contradiction readout (observe only — **downstream**).
3. **Intervene $\alpha=8$** tests a limited RepEng editor with controls — clears some false X, spares true contradictions (**downstream**).
4. **SAE features** decomposes the same **v** into Chanin L20 sparse features and runs top-k causal steers — ranking is not a labeled “temporality neuron” (**downstream**).
5. **Circuit C0-C2** (lab): L20 MLP is a causal **site** along **v**; locally sufficient — not a full circuit.
6. **Upstream** (:8258 tour): formation **d** @ L24 generalizes and Scores notes; ablate + $\alpha \cdot d$ are **null** — keep **d** and **v** separate.

Frozen story: Track A wrong_AUTO=0 (DEV+test). Downstream $\alpha=8$ partial editor + SAE pilot + L20 MLP site. Upstream map+gen YES / causal nulls — formation $\neq$ automatic commit editor.

Theory in one breath: concepts can live as directions in activation space; we **read** L20 (Deep dive), **write** along a frozen **v** at commit (Intervene), optionally **sparse-decompose** that write (SAE), locate a commit **MLP site** (C0-C2), and keep clinical decisions auditable at the symbolic boundary — while Upstream **d** stays a correlational formation map with honest nulls on causal tests.